MAT5210 Midterm Examination

1. Let $L : K$ be an algebraic field extension.
   (a) What does it mean for $L : K$ to be separable, normal, and Galois?

   Suppose $L : K$ is Galois.
   (a) Define the Galois group $\mathrm{Gal}(L/K)$.
   (b) Write down the fundamental theorem of Galois theory.

2. Let $\alpha := \sqrt{2+\sqrt{2}}$.
   (a) Find the minimal polynomial $m_\alpha(x)$ of α over $\mathbf{Q}$.
   (b) Let $L := \mathbf{Q}(\alpha)$. Show that K is a splitting field of $m_\alpha(x)$.
   (c) Find the Galois group $\mathrm{Gal}(L/\mathbf{Q})$.
   (d) Write down all the possible subfields of L under the Galois correspondence.

3. Let G be a finite group.
   (a) What does it mean for G to be solvable?
   (b) What does it mean for a polynomial $f(x) \in K[x]$ to be solvable by radicals over K?
   (c) Show that the polynomial $x^5 - p^2x - p$, where p is a prime number, is not solvable by radicals over $\mathbf{Q}$.